NATIONAL UNIVERSITY
COLLEGE OF COMPUTING AND
INFORMATION TECHNOLOGIES

# CCSFEN1L – INTRODUCTION TO SOFTWARE ENGINEERING

## Project Topic Proposal Form

**Section:** COM245 **Group Name:** Group Work ..

| **Proposed Project Title:** Online E-Print Service Management System |
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| Members: |
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## Project Overview:

The **E-Print Shop Management System** is a website that will allow students and customers to send their documents for printing online. The main purpose of this project is to make the printing process easier and faster, especially during busy days such as enrollment.

One of the main problems in the physical E-Print shop is the long line and overcrowding during peak days. Many students need to print important documents at the same time, which causes long waiting times and makes the shop crowded. Students may have to wait for a long time just to submit their documents for printing.

To solve this problem, we want to create a website where customers can upload their documents, choose their printing options, and place their orders online. After the order is processed and printed, the customer can simply walk in to the shop and pick up the printed documents.

The main goal of the project is to reduce the long lines and waiting time in the E-Print shop. It will also help the staff organize and manage printing orders more easily.

The system will include document uploading, printing option selection, price calculation, order submission, order status tracking, and order management for the shop staff. The system will focus on the online ordering and management of printing services.

## Objectives:

1. **To develop an E-Print Management System** that allows users to submit, manage, and monitor printing requests electronically.
2. **To provide an efficient printing request process** by reducing manual paperwork and minimizing the time required to process print requests.

3. **To enable administrators and authorized staff to manage printing requests**, including approving, updating, and tracking the status of requests.
4. **To provide users with real-time request status information** so they can easily determine whether their documents are pending, approved, printing, or completed.
5. **To maintain organized digital records of printing transactions** for easier monitoring, reporting, and future reference.
6. **To improve the accuracy and efficiency of print management** by reducing errors, duplicate requests, and miscommunication between users and printing personnel.

## Stakeholder and Target Users:

The primary stakeholders and target users of the E-Print Management System are **students, faculty, employees, printing staff, civilians and system administrators**.

- **Students and employees** will use the system to submit printing requests, upload documents, specify printing requirements, and monitor the status of their requests.
- **Faculty and staff** may use the system to request printing services for academic, administrative, or organizational purposes.
- **Printing staff** will manage submitted requests, review document details, process approved requests, update request statuses, and mark completed orders.
- **System administrators** will manage user accounts, monitor system activities, maintain records, and generate reports related to printing transactions.
- **Civilians** an use the system to submit printing requests, upload documents, specify printing requirements, and monitor the status of their requests. This allows members of the public to conveniently access printing services without relying entirely on manual procedures.

## Proposed System Features / Functional Requirements:

[List the major features or functions that the system is expected to provide]

1. No login/signup Required

**-Users can secure their print via paying online**

2. Document Upload -

**- -Support common file formats such as PDF, DOCX, and JPG/PNG**
**Users can upload documents for printing.**

3. Printing Request Form -

**- Number of copies**
**- Paper size (A4, Letter, Legal, etc.)**
**- Color or black-and-white**
**- Single-sided or double-sided**
**- Paper type**

**- Other special instructions**

4. Printing Cost Calculation -

**Users can see the estimated total before submitting their request.**

**The system automatically calculates the estimated printing cost based on the user's selected specifications.**

6. Admin/Printing Staff Dashboard -
**The administrator can view:**

**- Total users**
**- Total printing requests**
**- Pending requests**
**- Completed requests**
**- Cancelled requests**
**- Printing transactions**

7. Printing Staff Dashboard
**Printing staff can:**

**- View incoming requests**
**- Download submitted documents**
**- Review printing specifications**
**- Approve or reject requests**
**- Update request status**
**- Mark requests as completed**

8. User Notifications -
**Users can receive notifications when:**

**- Their request is submitted**
**- Their request is approved**
**- Their document is being printed**
**- Their document is ready for pickup**
- **- Their request is completed or rejected**

9. Printing History -
**Users can view their previous printing transactions, including:**

**- Document name**
**- Date of request**
**- Number of copies**
**- Printing specifications**
**- Total cost**

**- Request status**

10. Search and Filtering -

**Administrators and printing staff can search and filter requests by:**

**- User**
**- Request ID**
**- Date**
**- Status**
**- Document name**

11. Reports and Records -

**The administrator can generate reports such as:**

**- Daily/monthly printing requests**
**- Total printing transactions**
**- Completed and cancelled requests**
**- Estimated printing revenue**
**- Most frequently requested printing services**

## Non-Functional Requirements:

The proposed E-Print Shop Management System should meet the following quality requirements:

1. **Usability** – The website should be easy to use and understand, even for users who are not familiar with online printing systems.
2. **Performance** – The website should load pages and process orders within a reasonable amount of time, especially during busy days such as enrollment.
3. **Reliability** – The system should properly save printing orders and prevent orders from being lost during the process.
4. **Security** – User accounts and uploaded documents should be protected from unauthorized access.
5. **Availability** – The website should be available to customers whenever they need to submit their printing orders.
6. **Accuracy** – The system should correctly calculate the printing cost based on the selected printing options, such as number of pages, copies, and color or black-and-white printing.
7. **Maintainability** – The system should be easy to update and maintain when new features or changes are needed.
8. **Compatibility** – The website should work properly on common web browsers and different devices such as computers, tablets, and mobile phones.

## Proposed Technology Stack:

[Indicate the technologies that may be used to develop the system]

**Programming Language:** JavaScript / TypeScript
**Framework / Platform:** React.js with Node.js
**Database:** Mongodb **IDE/Development Environment:** Visual Studio Code
**Version Control System:** Git / GitHub
**Additional Libraries/Frameworks:** Express.js, Bootstrap or Tailwind CSS, and Axios

## Proposed Software Development Approach:

The proposed E-Print Management System will use an **Agile Software Development approach with MongoDB as the database management system**. Agile is suitable for the project because it allows the system to be developed in small and manageable stages. Each stage can be tested and improved based on feedback from the target users, including civilians, students, faculty, employees, printing staff, and administrators.

**MongoDB** will be used to store and manage the system's data, such as user accounts, printing requests, uploaded document information, printing specifications, request statuses, and transaction records. MongoDB is a **NoSQL document-oriented database**, which allows data to be stored in flexible document formats. This is useful for the E-Print Management System because different printing requests may contain different requirements, such as paper size, number of copies, color or black-and-white printing, and other specifications.

The development will be divided into several iterations. The first iteration will focus on user registration, login, and user account management. The second iteration will implement the printing request and document submission features. The following iterations will add request tracking, administrator and printing staff management, notifications, reports, and other necessary features. Each iteration will undergo testing and evaluation before moving to the next stage.

Using Agile together with MongoDB will allow the development team to **continuously improve the system, efficiently manage changing requirements, and provide a flexible database for storing printing-related information**.

## Proposed Evaluation:

The success of the E-Print Management System will be evaluated based on **functionality, usability, performance, reliability, and user satisfaction**.

- **Functionality:** Determine whether the system performs its intended functions, such as submitting, approving, processing, tracking, and recording printing requests.
- **Usability:** Evaluate whether users can easily understand and navigate the system without requiring extensive assistance.
- **Performance:** Assess the system's response time when submitting requests, uploading documents, viewing records, and updating request statuses.
- **Reliability:** Determine whether the system consistently stores and retrieves printing information accurately without significant errors or data loss.

- **User satisfaction:** Gather feedback from students, employees, printing staff, and administrators through questionnaires or surveys.
- **System testing:** Conduct functional and user acceptance testing to identify errors and verify that the system mee

## Expected Benefits / Impact:

The proposed E-Print Shop Management System is expected to provide several benefits to students, faculty, employees, civilians, printing staff, and administrators. The system is designed to make the printing process faster, more convenient, organized, and efficient by allowing customers to submit their printing requests online without needing to create an account or log in.

For students, employees, faculty, and civilians, the system will reduce the need to wait in long lines at the physical E-Print shop. Customers can upload their documents, select their printing specifications, view the estimated cost, and pay online before going to the shop. They can then monitor their order using their Order/Reference ID and visit the shop once their documents are ready for pickup. This will make the printing process more convenient and reduce unnecessary waiting time.

For printing staff, the system will provide an organized way to receive, review, and process printing requests. Staff can easily view uploaded documents, check printing specifications, verify payment status, and update the status of each order. This can help reduce errors, duplicate requests, misplaced documents, and miscommunication between customers and printing personnel.

For administrators, the system will provide centralized records of printing transactions and request information. Administrators can monitor printing activities, search and filter orders, track payments, and generate reports that can help in managing the E-Print service.

The removal of customer registration and login will also make the system more efficient and accessible. Customers will not need to create or remember an account before submitting a printing request. Instead, each order will be assigned a unique Order/Reference ID, while online payment will help verify and secure the transaction. Administrative and printing staff accounts will remain protected to ensure that only authorized personnel can access and manage internal system functions.

Overall, the proposed system is expected to reduce overcrowding and waiting time at the physical E-Print shop, improve the efficiency of printing operations, provide a more convenient customer experience, and maintain accurate digital records of printing transactions. The system can be especially beneficial during busy periods, such as enrollment, when the number of printing requests is significantly higher.

## Conclusion:

The **E-Print Shop Management System** is a website made to solve the problem of long lines and overcrowding at the physical E-Print shop, especially during busy days like enrollment. During these days, many students need to print their documents at the same time, causing students to wait for a long time before their documents can be

printed.

To solve this problem, the proposed system will allow students and other customers to submit their printing orders online. They can upload their documents, choose their printing options, and place their order through the website before going to the shop. The E-Print staff can then view and process the orders through the system. The main users of the system will be **students, customers, and E-Print shop staff**. Students and customers will use the website to send their documents and check the status of their orders, while the staff will use the system to manage and process the printing orders.

The expected benefit of the system is to **reduce the number of people waiting at the physical shop and make the printing process faster and more organized**. Instead of waiting in line to submit their documents, customers can order online and simply walk in to the shop when their documents are ready for pickup. This will also help the E-Print staff manage orders more easily, especially during busy days.

☐ **Approved**

☐ **Approved with Revisions**

☐ **Disapproved / Resubmission Required**

**Comments / Required Revisions:**

**Evaluated by:**

*Elsie V. Isip*
**Professor**